**Weather Application Prototype 3**

**Submitted by:** Sijan Kunwar

University id: 2551483

# INTERNET SOFTWARE ARCHITECTURE

## Weather App Summary

This is a browser-based weather application that was built with HTML, CSS, JavaScript, and PHP. It offers current weather information for any city the user inputs, based on data from the OpenWeatherMap API. Information on temperature, humidity, air pressure, wind speed, sunrise time, sunset time, and a short weather description is shown.

The application has a slim, responsive design that adjusts perfectly to screen sizes, providing a classy experience on both computers and mobile devices. Users can enter the name of a city to get weather information, which is stored locally as well with the help of localStorage so that it can be accessed offline. PHP scripts handle server-side

interactions with a MySQL database to save and fetch weather records instantly.

Advantages

Adaptive Design: The CSS design utilizes media queries to adapt the structure on different screen sizes, enhancing usability on any device.

Offline Functionality: Caching weather information in the browser's localStorage, the app allows users to view previously queried data offline.

Clean and Simple UI: The app uses a clean, minimalistic aesthetic combined with Font Awesome icons and a soft-focus background to create a visually pleasing interface.

Error Handling: The program provides error messages whenever users provide erroneous input or when the API face errors, and it makes users better understand problems.

Database Use: PHP handles proper storage and retrieval of weather information in a MySQL database with fewer frequent calls to the API.

Limitations

Security Flaws: Though the PHP script uses mysqli_real_escape_string to escape input, it lacks safer features like prepared statements and hence is open to SQL injection.

Generic Error Messages: Although the app does alert users to issues, the error messages are less informative and descriptive.

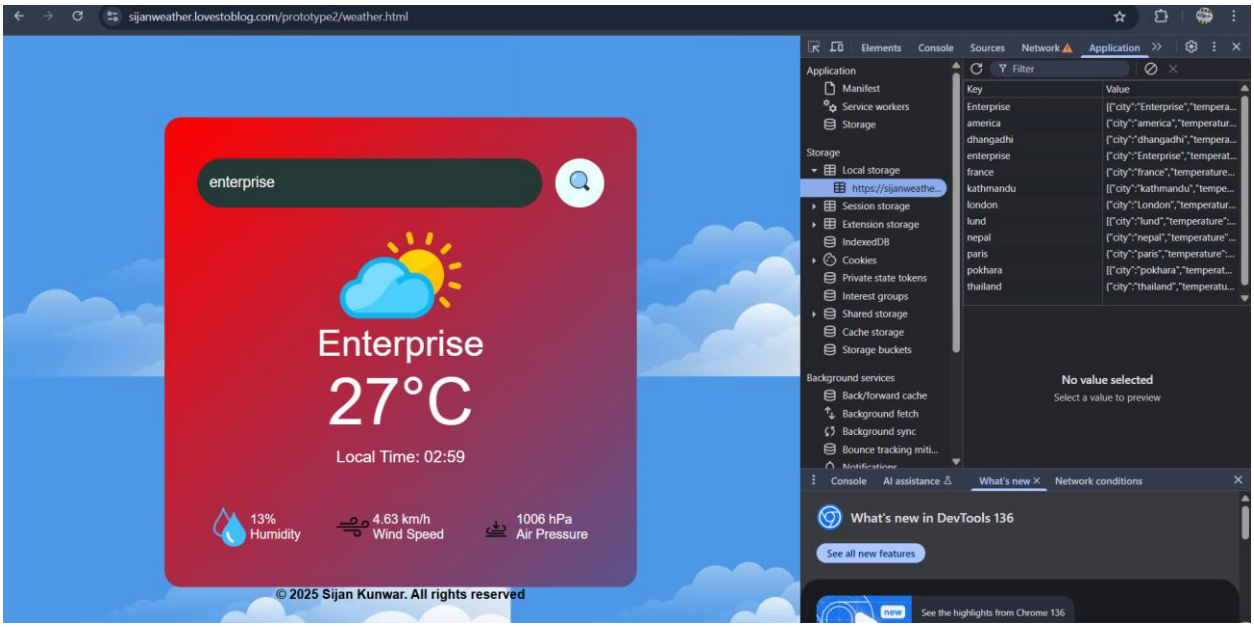
Dependence on API: The app is wholly reliant on the OpenWeatherMap API, and service downtime would make the app useless.

Exposed API Key: The API key in the PHP code is hardcoded, which is a security risk if accessed publicly.

Basic Offline Mode: The offline mode offers only viewing of stored data in an earlier state without offering any wider update facility or offline capability.

## Final Thoughts

On the whole, the weather app offers a smooth, intuitive experience with basic offline support and responsive UI. However, it could benefit from improved security practices, enhanced error reporting, and more extended offline capability to make it a more valuable and secure solution.

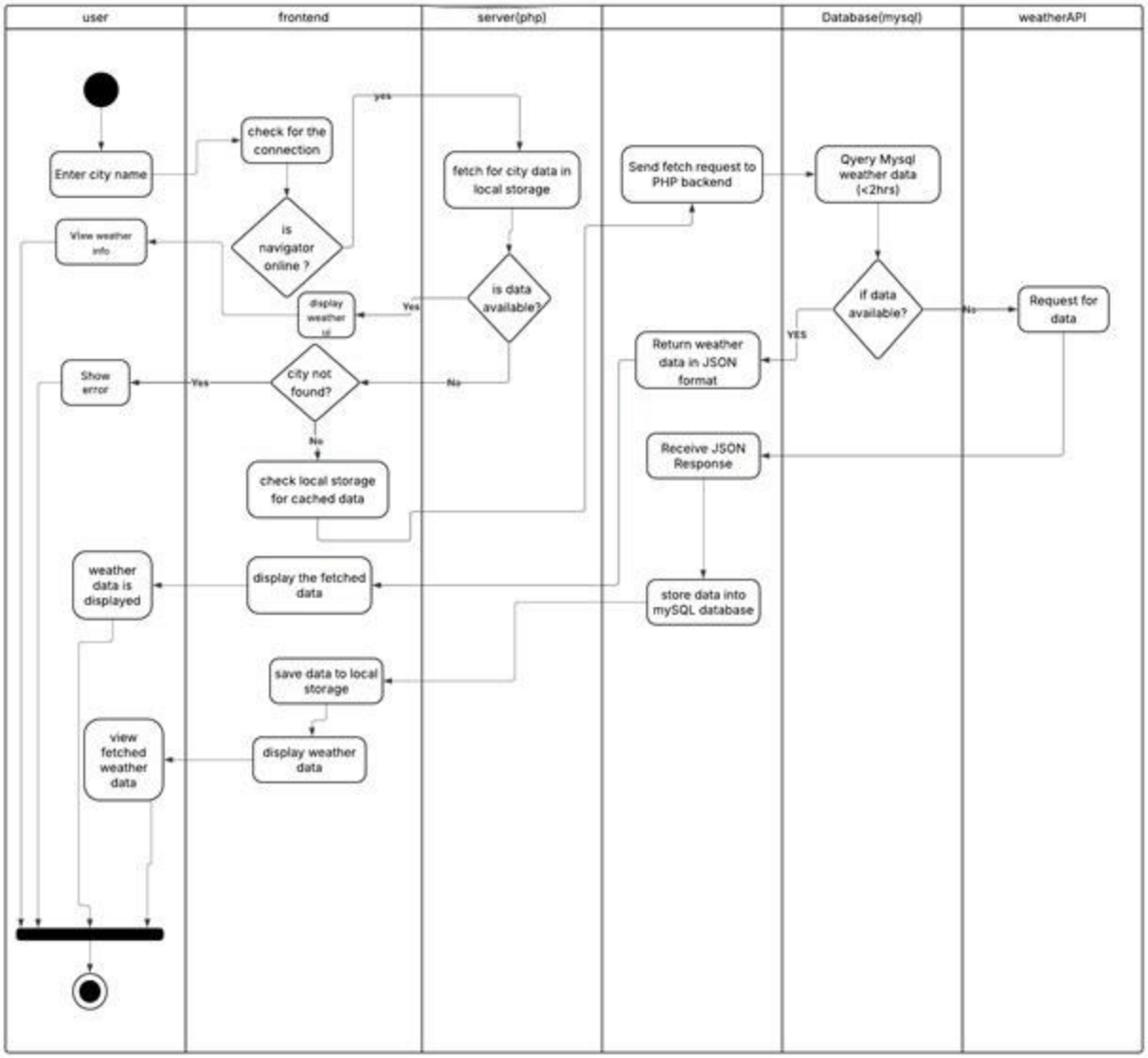



YouTube video link: https://www.youtube.com/watch?v=BcLKpJGSKtg

App link : https://sijanweather.lovestoblog.com/prototype2/weather.html

UML diagram:

1. Activity diagram

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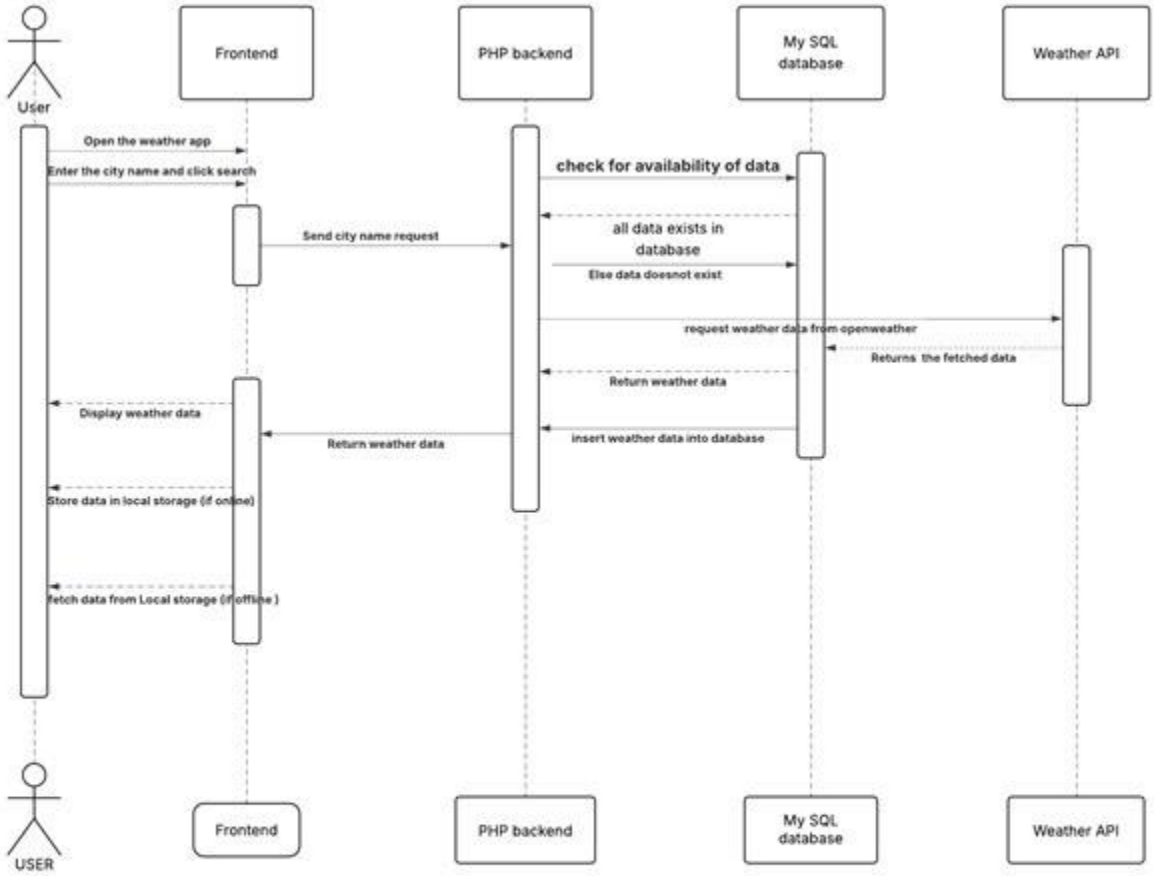
2. Sequence diagram

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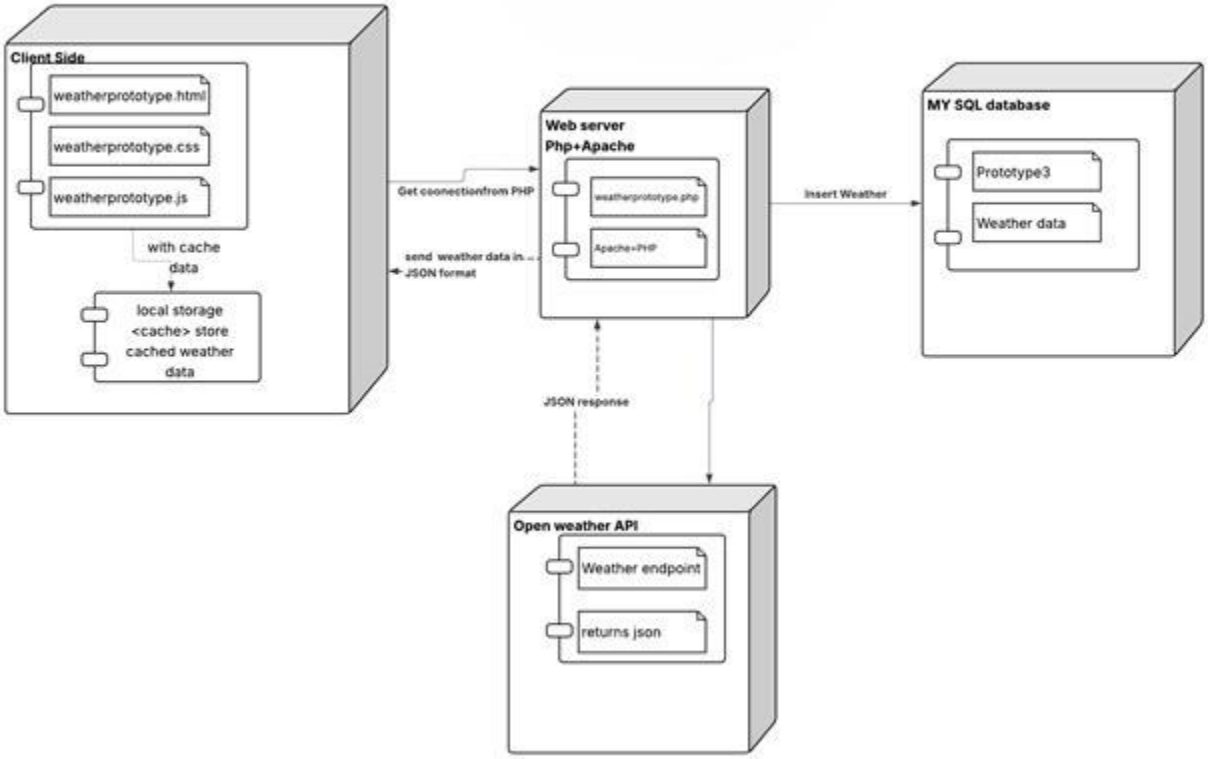
3. Deployment diagram

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